~~Package Insert~~

~~Brand or Product Name~~

Doxinyl 10%w/v ~~Doxycycline Hyclate~~ Oral Solution
Doxycycline Hyclate

Name and Strength of Active Substance(s)
1 ml contains: DOXYCYCLINE (HYCLATE), 100MG

Product Description
A yellow-brownish dense solution without precipitates or particles observable by simple visual inspection.

Pharmacodynamics
Doxycycline is a bacteriostatic agent that acts by interfering with the bacterial protein syntesis of sensitive species.

Doxycycline is a semi-synthetic tetracycline derived from oxytetracicline. It acts on the subunit 30S of the bacterial ribosome, to which is linked reversibly, blocking the union between aminoacyl-tRNA (transfer RNA) to the mRNA-ribosome complex, preventing the addition of new aminoacids into the growing peptide chain and thus interfering with protein synthesis.

Doxycycline is active against Gram-positive and Gram-negative bacteria.

Spectrum of activity:

Streptococcus spp.

Staphylococcus aureus

Chlamydia spp.

Mycoplasma spp.

Salmonella spp.

Pasteurella multocida

In vitro sensitivity of doxycycline against *Pasteurella multocida* and *Bordetella bronchiseptica* strains isolated from pigs has been determined by means of a plate diffusion method, and against *Mycoplasma hyopneumoniae* by a dilution method, with MIC_{90} values of 0.517 µg/ml, 0.053 µg/ml and 0.200 µg/ml, respectively.

According to the NCCLS standard, strains sensitive to doxycycline have MIC values below or equal to 4 µg/ml and those resistant have MIC values above or equal to 16 µg/ml.

There are at least two mechanisms of resistance to tetracyclines. The most important mechanism is due to decreased cellular accumulation of the drug. This is due to the establishment of either a pump elimination path or an alteration in the transport system that limits the uptake of tetracycline. The alteration in the transport system is produced by inducible proteins codified in plasmids and transposons.

The other mechanism is evidenced by decreased ribosome affinitiy for the Tetracycline-Mg^{2+} complex owing to chromosomal mutations. Resistance to tetracyclines may not only be the result of therapy with tetracyclines, but may also be caused by therapy with other antibiotics leading to selection of multi-resistant strains including tetracyclines. Although minimal inhibitory concentrations (MIC) tend to be lower for doxycycline than for older generation tetracyclines, pathogens resistant to one tetracycline are generally also resistant to doxycycline (cross resistance). Both long term treatment and treating for an insufficient length of time and/or sub-therapeutic dosages can select for antimicrobial resistance and should be avoided.

Pharmacokinetics

Doxycycline is bioavailable after oral administration. When orally administered, it reaches values greater than 70% in most species.
Feeding can modify the oral bioavailability of doxycycline. In fasting conditions bioavailability is around 10-15% greater than when the animal is fed.
Doxycycline is well distributed through the body as it is highly lipid soluble. It accumulated in liver, kidney, bones and intestine; enterohepatic recycling occurs. In lungs, it always reaches higher concentrations than in plasma. Therapeutic concentrations have been detected in aqueous humour, myocardium, reproductive tissues, brain and mammary gland. Plasma protein binding is 90-92%.
40% of the drug is metabolized and largely excreted through faeces (biliary and intestinal route), mainly as microbiologically inactive conjugates.

CHICKENS (broilers)
After oral administration, doxycycline is rapidly absorbed reaching maximum concentrations (C_{max}) around 1.5 h. Bioavailability was about 75%. The presence of food in the gastrointestinal tract reduces its absorption, reaching a bioavailability around 60% and extending considerably the time at which it reaches the maximum peak concentration, (t_{max}) 3.3 h.
PIGS
After an oral dose of 10 mg/kg/day, the concentration at steady state (Css) was around 1.30 µg/mL and plasma elimination half-life (t1/2) was 7.01 h.

Environmental Properties

Not relevant

Indication

CHICKENS (BROILERS)
Prevention and treatment of chronic respiratory disease (CRD) and mycoplasmosis caused by microorganisms sensitive to doxycycline.

PIGS
Prevention of clinical respiratory disease due to Pasteurella multocida and Mycoplasma hyopneumoniae sensitive to doxycycline.

The presence of the disease in the herd should be established before treatment.

Recommended Dosage and Mode of Administration

~~To be administered~~ Oral use in drinking water

CHICKEN (broilers): 11.5 – 23 mg doxycycline hyclate / kg body weight / day, corresponding to 0.1 - 0.2 ml of Doxinyl per kg body weight, for 3-5 consecutive days.

PIGS: 11.5 mg doxycycline hyclate/ kg body weight / day, corresponding to 0.1 ml of Doxinyl per kg body weight, for 5 consecutive days.

Based on the recommended dose, and the number and weight of the animals to be treated, the exact daily amount of Doxinyl should be calculated according to the following formula:

$$\frac{\text{X ml Doxinyl/ kg b.w./day} \times \text{Mean body weight (kg) of animals to be treated}}{\text{Mean daily water consumption (l) per animal}} = \text{X ml Doxinyl per l drinking water}$$

To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing. The uptake of medicated water is dependant on the clinical condition of the animals. In order to obtain the correct dosage, the concentration in drinking water may have to be adjusted.

The use of suitably calibrated weighing equipment is recommended if part packs are used. The daily amount is to be added to the drinking water such that all medication will be consumed in 24 hours. Medicated drinking water should be freshly prepared every 24 hours. It is recommended to prepare a concentrated pre-solution – approximately 100 grams product per litre drinking water – and to dilute this further to therapeutic concentrations if required. Alternatively, the concentrated solution can be used in a proportional water medicator.
Medicated water should be the only drinking source.
The remaining medicated water should be disposed of in accordance with local requirements.

If no improvement in clinical signs is seen within the treatment duration, the diagnosis should be reviewed and treatment changed.

~~Mode of Administration~~
~~Oral.~~

Contraindications
It is contraindicated in animals with known sensitivity to tetracyclines. Do not administer in animals with hepatic disturbances.

Warnings and Precautions
Not authorized for use in laying birds producing eggs for human consumption.

Interactions with Other Medicaments
Absoption of doxycycline may be diminished with the presence of calcium, iron, magnesium or aluminium in the diet. It is recommended to avoid administration in conjunction with antacids, kaolin and preparations containing iron.

Pregnancy and lactation
It is not recommended in animals during gestation and either during lactation.

Adverse Effects/Undesirable Effects
Allergic reactions and photosensitivity may appear, as for all tetracyclines.
If you notice any serious effects or other effects not mentioned in this package leaflet, please inform your veterinarian surgeon.

Overdose and Treatment
Not descriped.

Withdrawal Period(s)
Meat & offal: Chickens (broilers): 7 days

Pigs: 7 days.

Storage Conditions
Keep out of sight and reach of children. The medicinal product should be stored in its original package, protected from light at a temperature not above 30°C.

Shelf Life After First Opening

3 months.

Incompatibilities
Do not mix with other veterinary medicinal products.

Dosage Forms
Solution.

How to Dispose
Appropriately discard the empty container in a specified container.

Packaging Available
1L

Name and Address of Manufacturer
Industrial Veterinaria, S.A.
Esmeralda,19
E-08950 Esplugues de Llobregat (Barcelona) Spain.

Marketing Authorisation Holder
No 8, Jalan Astana 1F/KU2,
Bandar Bukit Raja, 41050,
Klang, Selangor DE. Malaysia

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~~November 12, 2018.~~
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